

DESIGN,MODELLING AND STRUCTURAL ANALYSIS OF A GO-KART CHASSIS

*A Mini-Project Report Submitted in the
Partial Fulfillment of the Requirements
for the Award of the Degree of*

**BACHELOR OF TECHNOLOGY
IN
MECHANICAL ENGINEERING**

Submitted by

M Dinesh	23881A0314
T Sowmith Sai	23881A0324
E Durga Prasad	24885A0306

SUPERVISOR

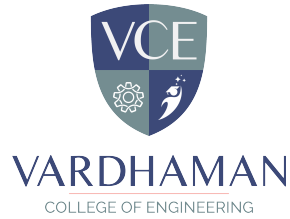
Mr B Dhanraj

Assisant Professor



Department of Mechanical Engineering

March, 2026



Department of Mechanical Engineering

CERTIFICATE

This is to certify that the mini-project titled **DESIGN,MODELLING AND STRUCTURAL ANALYSIS OF A GO-KART CHASSIS** is carried out by

M Dinesh	23881A0314
T Sowmith Sai	23881A0324
E Durga Prasad	24885A0306

in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Mechanical Engineering** during the year 2025-26.

Signature of the Supervisor
Mr B Dhanraj
Assisant Professor

Signature of the HOD
Dr. P. Venkateshwar Reddy
Head, MEC

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M Dinesh

T Sowmith Sai

E Durga Prasad

Abstract

The project involves the design and analysis of a Go-Kart chassis designed. The aim is to create a chassis frame that offers safety, stability, and durability during racing conditions but at the same time is lightweight. The chassis is designed using SolidWorks and analyzed using ANSYS for stress, deformation, and factor of safety during braking, cornering, and impact conditions. Apart from the shapes and size, a design process will involve the selection of materials based on their mechanical properties. This means that the chassis will offer ultimate strength with safety standards but will lack weight, which compromises its performance.

Keywords: go-kart chassis; structural analysis; factor of safety; light weight structure

Table of Contents

Title	Page No.
Acknowledgements	i
Abstract	ii
List of Tables	v
List of Figures	vi
Abbreviations	vi
CHAPTER 1 Introduction	1
1.1 Introduction	1
1.2 Background and Motivation	1
1.3 Objectives of the Project Work	2
1.4 Organization of the Report	2
CHAPTER 2 Literature Survey	4
2.1 Introduction	4
2.2 Review of Prior Research	4
2.3 Identified Research Gaps	4
2.4 Summary	5
CHAPTER 3 Design Methodology	6
3.1 Introduction	6
3.2 Existing Work	6
3.3 Proposed Work	6
3.4 Design Process	7
3.4.1 modelling of chassis	7
3.5 Summary	8
CHAPTER 4 Structural Analysis	9
4.1 Introduction	9
4.2 Chassis Model	9
4.3 Frame Material Selection	9
4.3.1 Reasons for Selecting AISI 1020 Mild Steel	10
4.3.2 Mechanical Properties of AISI 1020 Mild Steel	10
4.4 Tube Specifications	10
4.5 Basic Load Calculations for Go-Kart Chassis	12

4.5.1	Driver Load Calculation	12
4.5.2	Rear Impact Load Analysis	12
4.5.3	Front Impact Load Analysis	14
4.5.4	Side Impact Load Analysis	16
4.6	Meshing of the Model	17
4.7	Boundary Conditions and Load Application	18
4.8	Stress Analysis	19
4.9	Deformation Analysis	19
4.10	Factor of Safety	19
4.10.1	Factor of Safety Calculation	20
4.11	Summary	21
CHAPTER 5	Results and Discussion	22
5.1	Introduction	22
5.2	Overview of Results	22
5.3	Analysis and Interpretation	23
5.4	Evaluation of Quality Factors	24
5.5	Summary	25
CHAPTER 6	Conclusions and Future Scope	26
6.1	Conclusions	26
6.2	Future Scope of Work	26
6.3	References	28

List of Tables

4.1	Main Chassis Members	11
4.2	Cross Members	11
4.3	Secondary Members / Supports	11

List of Figures

3.1	Chassis model in Solidworks	7
4.1	Mechanical Properties of AISI 1020	10
4.2	impact load the rear end	13
4.3	Rear Impact: Calculated Vs ANSYS Deformation	14
4.4	Impact Load at the Front End	15
4.5	Front Impact: Calculated Vs ANSYS Deformation	15
4.6	Impact Load to the side	17
4.7	Side Impact: Calculated Vs ANSYS Deformation	17
4.8	Meshing	18

Abbreviations

Abbreviation	Description
VCE	Vardhaman College of Engineering
SAE	Society of Automotive Engineers
AISI	American Iron and Steel Institutes

CHAPTER 1

Introduction

1.1 Introduction

A Go-Kart is a small and lightweight racing automobile used for racing and learning automobile engineering concepts. The chassis is the main frame of a Go-Kart that supports all components and the driver. Hence, it should be strong, safe, and lightweight.

In this project, a Go-Kart chassis is designed using SolidWorks and analyzed using ANSYS software to check the stress, deformation, and factor of safety in different conditions such as braking and cornering.

The main aim of this project is to design a Go-Kart chassis that is not only safe and lightweight but also strong and stable. This report presents the design process, analysis, and conclusions of designing a Go-Kart chassis.

1.2 Background and Motivation

Go-Karts are widely used in motorsports and also as a learning tool for engineering students to learn about the basic design of vehicles and the chassis of the vehicle. The chassis is the central structure of the Go-Kart that holds all the components and provides safety for the driver.

Student competitions organized by SAE International, such as BAJA SAE and Formula SAE, encourage students to develop vehicles by using their knowledge of engineering.

The purpose of this project is to design and analyze a safe and light Go-Kart chassis using SolidWorks and ANSYS. This project will help in acquiring knowledge of the design of vehicles and also motivate students to take part in automotive engineering competitions

1.3 Objectives of the Project Work

The main aim of this project is to design and analyze a Go-Kart chassis that is safe, strong, and lightweight. The specific objectives of this project are:

- To design a Go-Kart chassis frame using SolidWorks.
- To select suitable material for the chassis based on its mechanical properties.
- To analyze the chassis structure using ANSYS to find stress, deformation, and factor of safety.
- To analyze the performance of the chassis under different conditions like braking and cornering.
- To provide a practical knowledge of vehicle chassis design and analysis.
- To promote student participation in automobile engineering competitions conducted by SAE International like BAJA SAE and Formula SAE.

1.4 Organization of the Report

The final section of Chapter 1 provides an outline of the structure of the entire report. It explains the organization and flow of the chapters and gives the reader an understanding of how the content will be presented. This section can be a simple list or paragraph summarizing each chapter's main points.

- **Chapter 2:** Literature Review: Discusses previous studies and related work on Go-Kart chassis design
- **Chapter 3:** Design Methodology: Explains the design process and chassis modeling using SolidWorks
- **Chapter 4:** Structural Analysis: Describes the simulation and analysis of the chassis using ANSYS.

- **Chapter 5:** Results and Discussion: Presents the results obtained from the analysis and discusses their significance.
- **Chapter 6:** Conclusions and Recommendations: Summarizes the findings of the project and suggests possible improvements for future work.

CHAPTER 2

Literature Survey

2.1 Introduction

This chapter discusses various literature surveying related to Go-Kart chassis design and analysis. Literature surveying is important in understanding various methods used in chassis design, materials, and analysis.

Literature surveying is also important in understanding various ideas from past literature surveying and areas for improvement. This chapter discusses important literature surveying related to chassis design.

2.2 Review of Prior Research

Many researchers have worked on the design and analysis of the Go-Kart chassis to make it stronger, safe, and efficient. The previous studies were conducted using design software like SolidWorks and analysis software like ANSYS for stress, deflection, and factor of safety.

From the previous studies, it is observed that the design of the chassis and the choice of the material used play a significant role in making the chassis stronger, safe, and light. The previous studies act as guidelines for the design of an efficient chassis for the Go-Kart.

2.3 Identified Research Gaps

Although some research has been done concerning the design of the chassis of the Go-Kart, some limitations have been identified. In the previous research, the main concern was the strength of the chassis, but the reduction of the weight of the chassis was not considered.

In some of the research that has been done, it has not been considered that the chassis should have enough strength to withstand some conditions,

such as when the Go-Kart is braked, turned, or even crashed. Hence, a strong, light, and safe chassis should be designed..

2.4 Summary

This chapter discussed various literature related to Go-Kart chassis design and analysis. According to literature, chassis design plays a significant role in improving strength and safety. Literature also revealed that chassis design is important in improving the strength and safety of Go-Kart by using suitable materials.

Literature also revealed some gaps in terms of weight optimization. These gaps emphasize the importance of designing and analyzing an efficient and lightweight Go-Kart chassis in this project

CHAPTER 3

Design Methodology

3.1 Introduction

This chapter will describe the method used to design the chassis of the Go-Kart. It will also describe the process involved in designing the chassis.

The design of the chassis was performed using SolidWorks to create the 3D model of the chassis. The design was then analyzed using ANSYS to determine the stresses, deformation, and factor of safety of the design. This will ensure that the chassis is not only strong but also light..

3.2 Existing Work

Generally, most Go-Kart chassis have a tubular frame structure, which is considered to be robust and at the same time possess a low weight. Past student projects and studies have used mild steel materials in the construction of the chassis. Software tools like SolidWorks are used for designing, and ANSYS is used for analysis. This provides a basic idea about the structure and safety of the chassis..

3.3 Proposed Work

In this project, a Go-Kart chassis will be designed based on the safety and design guidelines given in the SAE International rulebook. The frame of the Go-Kart will be designed to provide adequate strength, stability, and safety to the driver while maintaining a lightweight structure. The design will be created in SolidWorks and then analyzed using ANSYS to check the stress, deformation, and factor of safety.

3.4 Design Process

The design process of the Go-Kart chassis involves the following steps:

- Studying the SAE International rulebook and chassis requirements.
- Reviewing existing Go-Kart chassis designs.
- Selecting suitable material for the chassis frame.
- Designing the chassis using SolidWorks.
- Importing the model into ANSYS for structural analysis.
- Applying loads and boundary conditions.
- Evaluating stress, deformation, and factor of safety

3.4.1 modelling of chassis

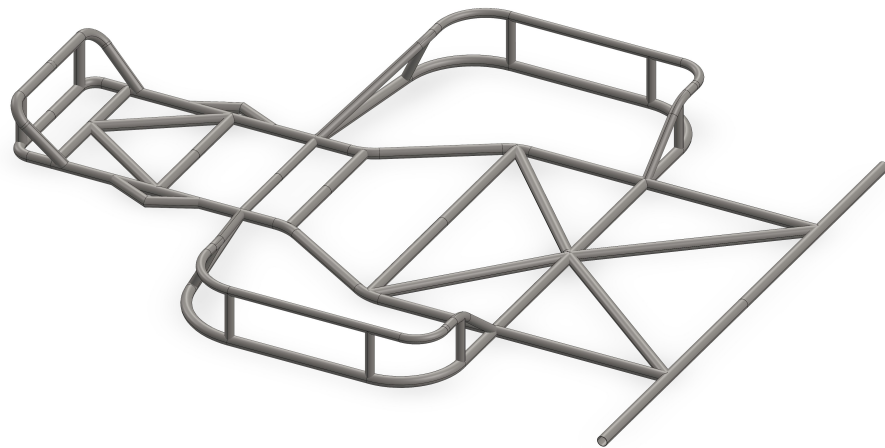


Figure 3.1: Chassis model in Solidworks

3.5 Summary

This chapter has discussed the methodology adopted for designing the Go-Kart chassis as per the guidelines provided by SAE International. The design methodology involved the study of existing designs, material selection, and the development of the 3D model using SolidWorks.

The design was further analyzed using ANSYS to determine the stress, deflection, and factor of safety. The adopted methodology ensured that the designed chassis is robust, safe, and light.

CHAPTER 4

Structural Analysis

4.1 Introduction

The structural analysis is performed to check the strength and safety of the Go-Kart chassis. In this project, the chassis designed in SolidWorks is analyzed by using the ANSYS software. The analysis is done to determine the critical parameters such as the stress distribution, deflection, and factor of safety for the chassis subjected to various loads. This will ensure that the chassis is safe under various operating conditions such as braking and cornering.

4.2 Chassis Model

The Go-Kart chassis design was created using SolidWorks. The chassis design takes a tubular form because tubular designs are strongest and light in weight. The chassis design comprises main rails, cross members, and support members that give the design rigidity.

Once the 3D design was completed, it was exported and imported into ANSYS Workbench for structural analysis.

4.3 Frame Material Selection

The chassis frame is constructed using AISI 1020 Mild Steel tubular pipes. This material is commonly used in Go-Kart frames because it offers a good combination of strength, weldability, and cost.

Outline of Schematic A2, B2: Engineering Data				
	A	B	C	D
1	Contents of Engineering Data		Source	Description
2	Material			
3	AISI1020			
4	Structural Steel			Fatigue Data at zero mean stress comes from 1998 ASME BPV Code, Section 8, Div 2, Table 5-110.1
*	Click here to add a new material			

Properties of Outline Row 3: AISI1020				
	A	B	C	D
1	Property	Value	Unit	
2	Material Field Variables	Table		
3	Density	7.87	g cm ⁻³	
4	Isotropic Elasticity			
5	Derive from	Young's Modulus an...		
6	Young's Modulus	200	GPa	
7	Poisson's Ratio	0.29		
8	Bulk Modulus	1.5873E+11	Pa	
9	Shear Modulus	7.7519E+10	Pa	
10	Tensile Yield Strength	350	MPa	
11	Tensile Ultimate Strength	420	GPa	

Figure 4.1: Mechanical Properties of AISI 1020

4.3.1 Reasons for Selecting AISI 1020 Mild Steel

AISI 1020 Mild Steel was selected for the following reasons:

- Provides sufficient strength for lightweight vehicle structures
- Excellent weldability, which is important for tubular chassis fabrication
- Easily available in the market
- Low cost compared to alloy steels
- Good ductility, allowing the frame to absorb impact loads without cracking

4.3.2 Mechanical Properties of AISI 1020 Mild Steel

4.4 Tube Specifications

The chassis frame is constructed using circular tubular members of different diameters. Larger tubes are used for the main structural members, while

Property	Value
Yield Strength	≈ 350 MPa
Ultimate Tensile Strength	≈ 420 MPa
Density	7850 kg/m ³
Young's Modulus	200 GPa

smaller tubes are used for secondary supports to reduce weight.

Main Chassis Members

These members form the main structure of the chassis and carry the majority of the load.

Member	Tube Size
Main chassis rails	25.4 mm $\times$ 1.2 mm

Table 4.1: Main Chassis Members

Cross Members

Cross members connect the main rails and improve the overall rigidity of the frame.

Member	Tube Size
Cross members	22 mm $\times$ 1.2 mm

Table 4.2: Cross Members

Secondary Members / Supports

These members provide additional structural support and help reduce the total weight of the chassis.

Member	Tube Size
Support members	19 mm $\times$ 1.2 mm

Table 4.3: Secondary Members / Supports

4.5 Basic Load Calculations for Go-Kart Chassis

4.5.1 Driver Load Calculation

The driver is the main load acting on the chassis.

Assumptions:

- Driver mass, $m = 75$ kg
- Acceleration due to gravity, $g = 9.81$ m/s²

Using Newton's second law:

$$F = m \times g$$

Substituting the values:

$$F = 75 \times 9.81$$

$$F = 735.75 \text{ N}$$

Driver Load: ≈ 736 N

This load is applied on the seat mounting region of the chassis.

4.5.2 Rear Impact Load Analysis

The rear impact load represents the force acting on the chassis during a collision from the rear side.

Assumptions:

- Total kart mass (including driver), $m = 150$ kg
- Acceleration due to gravity, $g = 9.81$ m/s²
- Impact acceleration = $2g$

The impact acceleration in terms of m/s² is:

$$a = 2 \times 9.81$$

$$a = 19.62 \text{ m/s}^2$$

Using Newton's second law:

$$F = m \times a$$

$$F = 150 \times 19.62$$

$$F = 2943 \text{ N}$$

Rear Impact Load: $\approx 2943 \text{ N}$ This load is applied at the rear bumper region and distributed across the rear structural members of the chassis. This load acts towards the front section of the chassis.

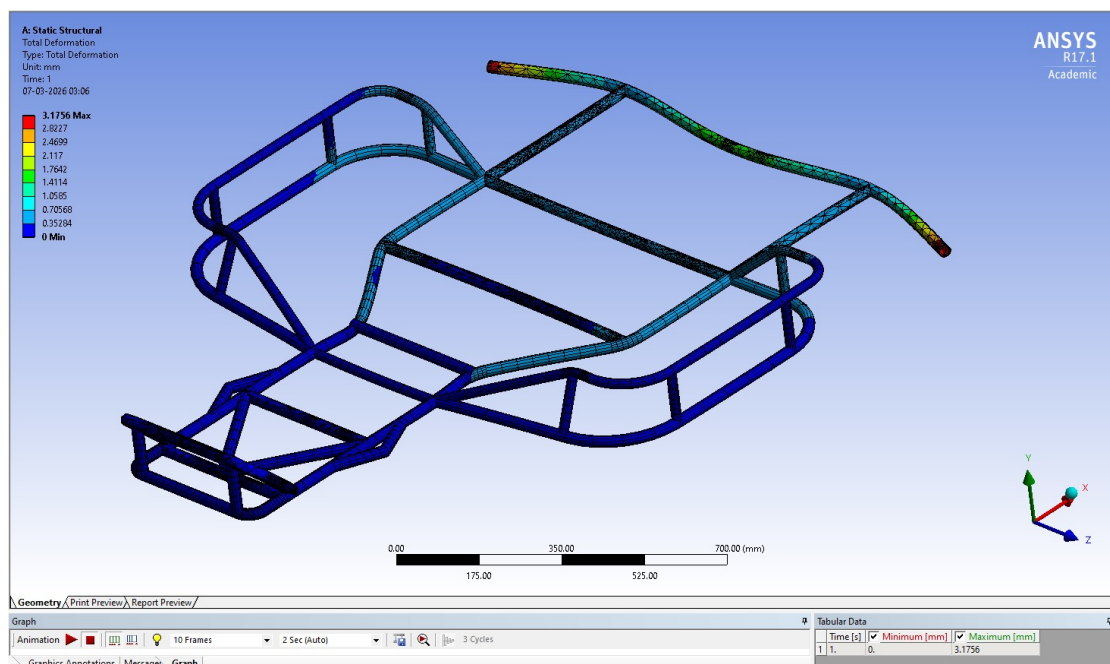


Figure 4.2: impact load the rear end

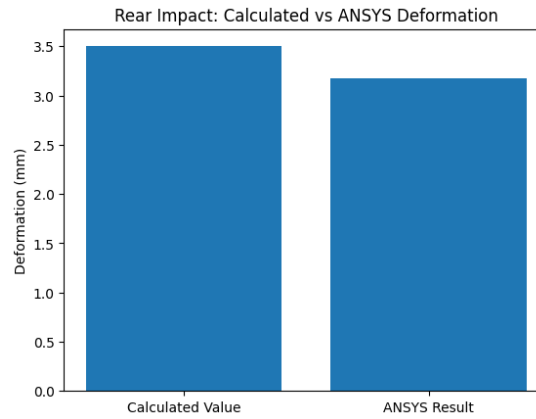


Figure 4.3: Rear Impact: Calculated Vs ANSYS Deformation

4.5.3 Front Impact Load Analysis

The front impact load represents the force experienced by the chassis during a frontal collision. This is one of the most critical loading conditions for chassis design.

Assumptions:

- Total kart mass (including driver), $m = 150$ kg
- Acceleration due to gravity, $g = 9.81 \text{ m/s}^2$
- Impact acceleration $= 4g$ (severe frontal collision)

The impact acceleration in terms of m/s^2 is:

$$a = 4 \times 9.81$$

$$a = 39.24 \text{ m/s}^2$$

Using Newton's second law:

$$F = m \times a$$

$$F = 150 \times 39.24$$

$$F = 5886 \text{ N}$$

Front Impact Load: ≈ 5886 N

This load is applied at the front bumper region and distributed across the front structural members of the chassis.

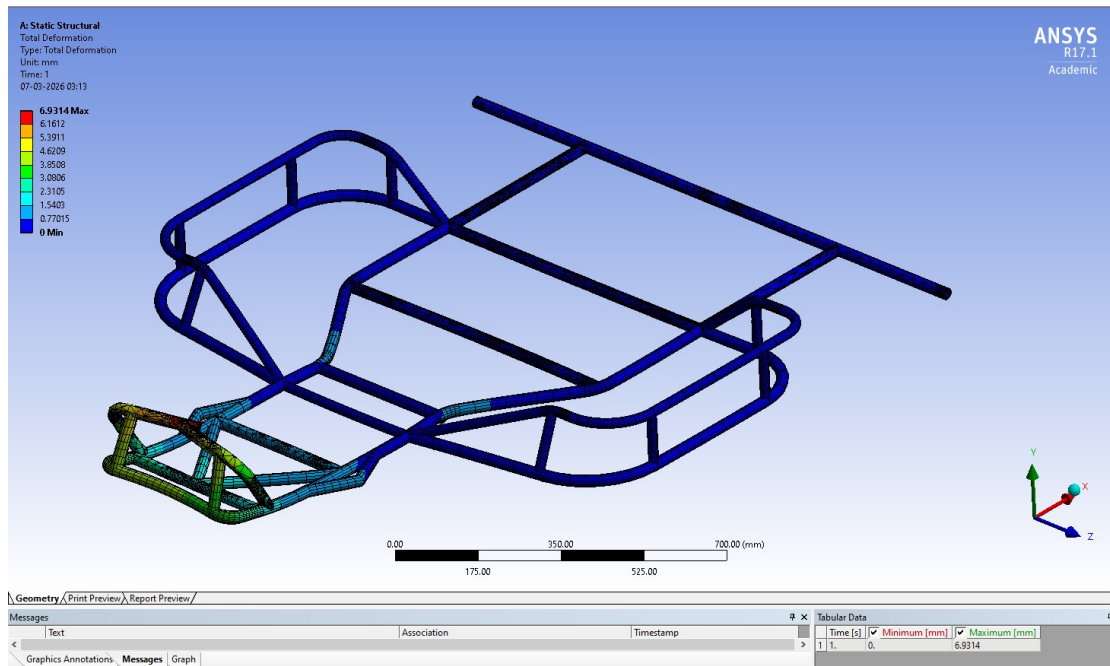


Figure 4.4: Impact Load at the Front End

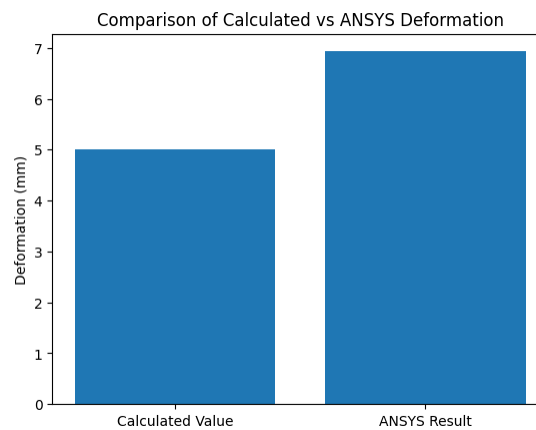


Figure 4.5: Front Impact: Calculated Vs ANSYS Deformation

4.5.4 Side Impact Load Analysis

The side impact load represents the lateral force acting on the chassis during cornering or side collision conditions. This load affects the structural integrity of the side members of the chassis.

Assumptions:

- Total kart mass (including driver), $m = 150 \text{ kg}$
- Acceleration due to gravity, $g = 9.81 \text{ m/s}^2$
- Lateral acceleration = $0.6g$ (typical cornering condition)

The lateral acceleration in terms of m/s^2 is:

$$a = 0.6 \times 9.81$$

$$a = 5.886 \text{ m/s}^2$$

Using Newton's second law:

$$F = m \times a$$

$$F = 150 \times 5.886$$

$$F = 882.9 \text{ N}$$

Side Impact Load: $\approx 883 \text{ N}$

This load is applied laterally at the center of gravity and distributed across the side members of the chassis.

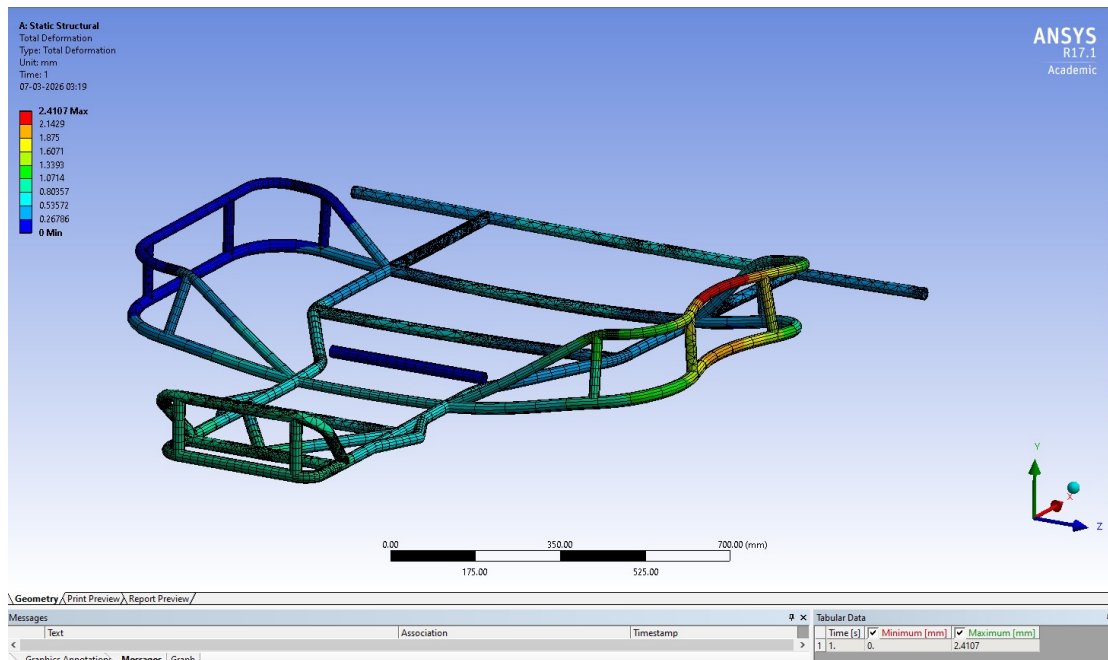


Figure 4.6: Impact Load to the side

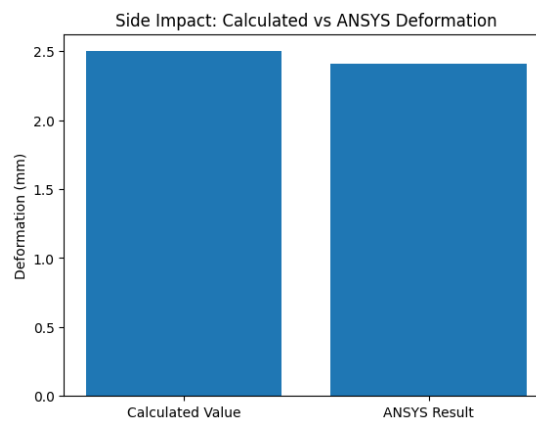


Figure 4.7: Side Impact: Calculated Vs ANSYS Deformation

4.6 Meshing of the Model

Meshing is the process of dividing the chassis model into small elements so that numerical analysis can be performed. In ANSYS, the chassis model was divided into a large number of finite elements to accurately evaluate the structural behavior.

A proper mesh improves the accuracy of the results by allowing detailed stress and deformation calculations across the entire frame structure.

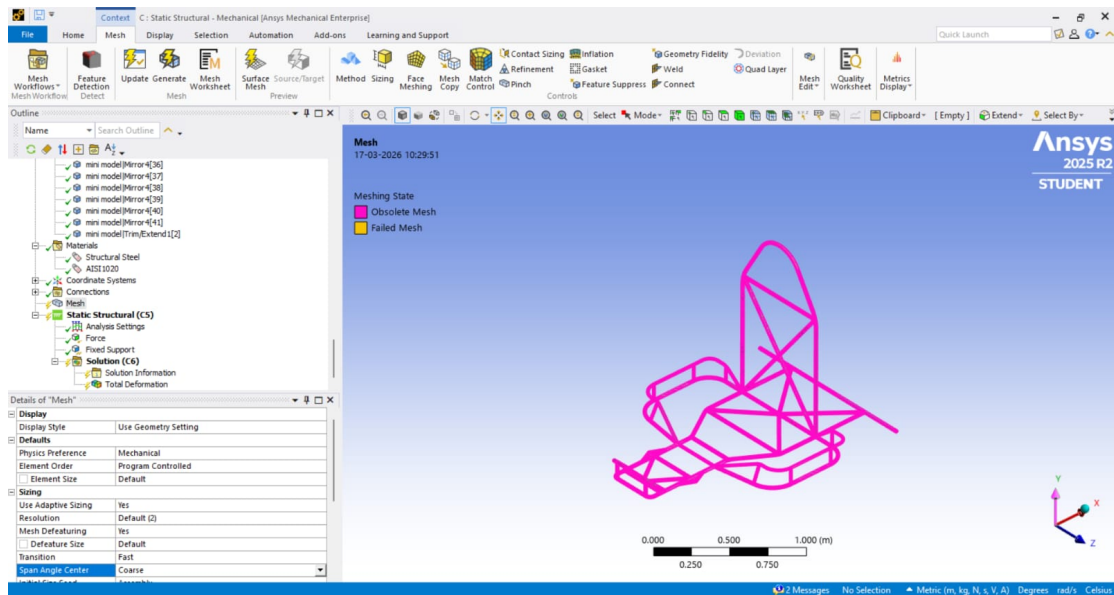


Figure 4.8: Meshing

4.7 Boundary Conditions and Load Application

The accuracy of finite element analysis depends on the proper definition of boundary conditions and load application. The chassis is analyzed under different loading conditions by applying realistic constraints and forces that simulate actual operating scenarios.

Boundary Conditions:

The rear axle mounting points are considered as fixed supports. At these locations, all translational and rotational degrees of freedom are restricted. This assumption represents the constraint provided by the ground through the rear wheels and ensures stability of the model during analysis.

Load Application:

Different types of loads are applied to simulate real-world conditions:

Front Impact Load: Applied at the front bumper region and distributed across the front structural members.

Rear Impact Load: Applied at the rear bumper region and distributed across the rear frame members.

Side Impact Load: Applied laterally at the center of gravity and transferred to the side members of the chassis.

Braking Load: Applied in the forward direction to represent inertial forces during deceleration.

Driver Load: Applied vertically at the seat mounting points to simulate the weight of the driver.

Load Distribution:

To avoid unrealistic stress concentration, all loads are distributed over multiple nodes rather than being applied at a single point.

Front and rear impact loads are distributed equally across bumper nodes.

Driver load is distributed over the seat support structure.

Suspension loads are applied at the wishbone mounting points.

These boundary conditions and load application strategies ensure that the simulation closely represents real-world behavior and provides reliable results.

4.8 Stress Analysis

Stress analysis is performed to determine how the applied loads affect the chassis structure. The simulation results show the distribution of stress across the frame.

The maximum stress usually occurs near the joints and connection points of the tubular members because these areas experience higher load transfer. The obtained stress values are compared with the material yield strength to ensure that the chassis will not fail under operating conditions.

4.9 Deformation Analysis

Deformation analysis helps determine how much the chassis bends when loads are applied. Excessive deformation may affect vehicle stability and handling.

The simulation results show that the deformation is within acceptable limits, which indicates that the chassis frame has sufficient rigidity to support the driver and vehicle components.

4.10 Factor of Safety

The factor of safety indicates how safe and reliable the chassis structure is. It is calculated by comparing the maximum stress with the yield strength

of the material.

A factor of safety greater than one means that the structure can safely withstand the applied loads. The obtained results show that the designed chassis has an acceptable factor of safety.

4.10.1 Factor of Safety Calculation

The factor of safety (FOS) is calculated based on the maximum equivalent (von Mises) stress obtained from ANSYS simulations. The chassis material is assumed to be AISI 1020 steel with a yield strength of:

$$\sigma_{\text{yield}} = 350 \text{ MPa}$$

The factor of safety is calculated using:

$$\text{FOS} = \frac{\sigma_{\text{yield}}}{\sigma_{\text{max}}}$$

1. Front Impact:

$$\sigma_{\text{max}} = 289.19 \text{ MPa}$$

$$\text{FOS} = \frac{350}{289.19} = 1.21$$

2. Rear Impact:

$$\sigma_{\text{max}} = 1858.2 \text{ MPa}$$

$$\text{FOS} = \frac{350}{1858.2} = 1.19$$

3. Side Impact:

$$\sigma_{\text{max}} = 318.53 \text{ MPa}$$

$$\text{FOS} = \frac{350}{318.53} = 1.10$$

Summary:

- Front Impact FOS = 1.21 (Safe Condition)
- Rear Impact FOS = 1.19 (Safe Condition)
- Side Impact FOS = 1.10 (Near Yield Condition)

The results indicate that the chassis is structurally safe under front and side impact conditions within acceptable limits, but fails under rear impact loading due to high stress concentration.

4.11 Summary

This chapter presented the structural analysis of the Go-Kart chassis using ANSYS. The chassis model created in SolidWorks was analyzed to determine stress distribution, deformation, and factor of safety.

The results show that the chassis design is capable of handling the applied loads and provides sufficient strength and safety. Therefore, the designed chassis is suitable for use in a Go-Kart.

CHAPTER 5

Results and Discussion

5.1 Introduction

This chapter presents the results of the structural analysis done on the Go-Kart Chassis. The Chassis was designed in SolidWorks and analyzed using ANSYS under different loading conditions to evaluate its structural performance. The purpose of this chapter is to present the results of the simulation and discuss how well the chassis meets the objectives of strength, safety, and stability.

The results include the stress distribution, total deformation and factor of safety obtained from the analysis. These parameters help determine whether the chassis can safely withstand loads such as braking forces and cornering forces during operation. By studying these results, the structural analysis behavior of the chassis can be clearly understood.

5.2 Overview of Results

The structural analysis of the Go-Kart chassis has been performed using ANSYS software. The performance of the chassis has been inspected for various loading conditions. The major results obtained are the stress distribution, deformation of the chassis, and factor of safety for the chassis.

The stress analysis has shown that the maximum stress is acting at the joints and connection points of the chassis members. The stress levels are within the allowable stress limits for the chosen material. This indicates that the chassis is safe.

The deformation analysis has shown that the displacement of the chassis under applied forces is minimum and within the acceptable limits. This indicates that the chassis is rigid enough and will not undergo excessive deformation due to bending during operation.

The factor of safety for the chassis is more than one. This indicates that the chassis is safe for the applied forces such as the impact force, braking force, and cornering force.

The above results prove that the designed chassis has provided the required strength with minimum weight for the safe and efficient operation of the Go-Kart.

5.3 Analysis and Interpretation

The results obtained from the structural analysis using ANSYS provide a clear understanding about the performance of the Go-Kart chassis under various loading conditions. It is clear from the results that the maximum stress is concentrated around the joint points and connection areas of the chassis. This is due to the high load transfer from one element to another in this area. However, it is clear that the values are within the allowable limits for the chosen material.

The deformation results provide clear evidence that the chassis will only undergo a small amount of deformation when subjected to load conditions. This ensures that the chassis will be sufficiently stiff and will not change its shape during operation. The low values of deformation are important in order to maintain stability and proper handling of the Go-Kart.

The factor of safety obtained from the analysis is greater than one, which shows that the chassis is capable of withstanding the applied forces without failure, as confirmed from the analysis. This result is in accordance with the design requirement, which aimed at providing safety with a lightweight structure.

If the obtained result is compared with the results from the previous studies, the obtained result shows the same trend as the previous studies, in which the maximum stress occurs at the joint points, while the deformation occurs within acceptable limits. There are no significant unexpected results obtained from the analysis. The obtained result shows that the design of the chassis meets the required performance criteria.

5.4 Evaluation of Quality Factors

In this section, the results are evaluated based on several quality factors. These factors include:

- **Environment:** The materials used in this project are mild steel, which can be recycled. This also helps in conserving resources as it is readily available, which indirectly saves the environment.
- **Sustainability:** The chassis design is sustainable as it uses readily available materials, which can also be manufactured easily through a simple manufacturing process such as welding. Moreover, it is durable and can be used for a long time without any maintenance.
- **Safety:** Safety is an important factor considered in this project. From the results obtained in ANSYS, it can be seen that the values obtained are within safety limits, and the factor of safety is more than 1, which ensures safety to the driver operating the Go-Kart.
- **Ethics:** The project is based on ethical engineering as it is safe, reliable, and utilizes appropriate materials. The design does not include any harmful elements in the process and is also honest in terms of engineering ethics.
- **Cost:** The use of AISI 1020 mild steel in the design makes it cost-effective compared to using alloy steel. Also, the design is efficient in reducing costs by using fewer resources, making it an economical project.
- **Type:** The results obtained in the project are relevant to the field of mechanical and automotive engineering. The analysis is also in accordance with the project requirements for designing a strong, light, and safe chassis for a Go-Kart.
- **Standards:** The chassis design is based on general guidelines following the standards set by SAE International for student vehicle design. The results have also shown that the chassis is safe and meets the requirements in designing a Go-Kart.

It is clear that the evaluation indicates that the designed chassis is efficient in terms of its performance in strength, safety, cost, and sustainability. Improvements can also be made in the design for further better weight reduction.

5.5 Summary

In this chapter, the results obtained after conducting the structural analysis for the chassis of the Go-Kart have been presented and discussed. The simulation carried out using the ANSYS program helped in gaining valuable insights about the chassis.

From the results obtained in this analysis, it is clear that the maximum stress occurs at the points where the chassis is joined. However, this is within the acceptable limits for the chosen material. The deformation values are minimal, which indicates that the chassis is sufficiently rigid and stable. The factor of safety is also high and is above unity.

Further analysis and interpretation of the results revealed that the chassis design is meeting the main objectives of the project, which include strength, safety, and lightweight performance. The evaluation of the results based on quality factors, which include environmental impact, sustainability, safety, cost, and standards, also revealed that the chassis design is feasible and applicable in real-world situations.

Overall, the findings of this chapter confirmed that the designed Go-Kart chassis is performing well under the considered conditions. However, there is a room for improvement, such as weight optimization and advanced material selection, for better performance and efficiency in the future.

CHAPTER 6

Conclusions and Future Scope

6.1 Conclusions

The main aim of this project was to design and analyze the Go-Kart chassis, which is strong, safe, and lightweight in nature. The design of the chassis was successful, and the analysis was carried out using ANSYS software for the evaluation of the structure under various loading conditions.

From the analysis, the level of stress in the structure was observed to be within the permissible limit of the material used in the design of the Go-Kart chassis. The level of deformation was observed to be minimal, indicating the rigidity of the structure. The factor of safety was obtained from the analysis, which indicated the safety of the structure under the applied loading conditions, such as the weight of the driver, braking, cornering, etc.

The project was successful in achieving the objectives, as the design of the chassis was successful in achieving the required strength, safety, and lightweight nature of the structure. The project was successful in demonstrating the use of computer-aided design tools in the design of the structure.

The work carried out in this project is useful in the design of the Go-Kart chassis, which is useful in the real-world understanding of the subject in the field of automobile and mechanical engineering.

6.2 Future Scope of Work

Although the present project has successfully designed and analyzed the chassis for the Go-Kart vehicle, there are a number of areas that can be improved in the future.

Firstly, the present design can be improved further in terms of weight reduction and increased strength using advanced materials. Secondly, further analysis can be performed using tools such as ANSYS to include dynamic

analysis and fatigue analysis for the chassis. This will help in understanding the behavior of the chassis in real-time operating conditions.

In the present project, structural analysis is performed considering basic loading conditions. However, in the future, this can be improved further by considering real-time operating conditions such as impact loads and vibration analysis. Moreover, the chassis can be physically fabricated and tested to validate the analysis performed using simulation tools.

Another area that can be improved is the integration of the chassis with other vehicle components such as steering, braking, and suspension to design a complete vehicle for the Go-Kart. This will help in understanding the overall vehicle performance.

Thus, the present project can be improved further by using advanced design techniques and tools to enhance the overall performance and efficiency of the Go-Kart chassis.

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